

**CS1231S: Discrete Structures**  
**Tutorial #5: Relations & Partial Orders**  
(Week 7: 4 – 8 March 2024)

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**I. Discussion Questions**

- D1. Let  $R$  be a binary relation on a non-empty set  $A$ . If  $R = \emptyset$ , then is  $R$  reflexive? Symmetric? Transitive?
- D2. Suppose a binary relation  $R$  on a non-empty set  $A$  is reflexive, transitive, symmetric and antisymmetric. What can you conclude about  $R$ ? Explain.
- D3. Asymmetry is defined in question 6 as follows: A binary relation  $R$  on a set  $A$  is **asymmetric** iff  
$$\forall x, y \in A (x R y \Rightarrow y \not R x).$$
  
Are there binary relations that are both symmetric and asymmetric?
- D4. Let  $R$  be a relation on  $A$ . Prove that the following is an alternative definition of **antisymmetry**:  
$$\forall x, y \in A \left( (x \neq y) \Rightarrow \left( ((x, y) \in R) \Rightarrow ((y, x) \notin R) \right) \right).$$

**II. Tutorial Questions**

1. Let  $S$  be the set of all strings over the alphabet  $\mathcal{A} = \{s, u\}$ , i.e. an element of  $S$  is a sequence of characters, each of which is either  $s$  or  $u$ . Examples of elements of  $S$  are:  $\varepsilon$  (the empty string),  $s, u, sus, usssuu$ , and  $susussssu$ .  
Define a relation  $R$  on  $S$  by the following:  $\forall a, b \in S (a R b \Leftrightarrow \text{len}(a) \leq \text{len}(b))$   
where  $\text{len}(x)$  denotes the length of  $x$ , i.e. the number of characters in  $x$ .  
Is  $R$  antisymmetric? Prove or disprove it.
2. Consider the “divides” relation on each of the following sets of integers. For each of them, draw a Hasse diagram, and find all minimal, maximal, smallest and largest elements.  
(a)  $A = \{1, 2, 4, 5, 10, 15, 20\}$ .  
(b)  $B = \{2, 3, 4, 6, 8, 9, 12, 18\}$ .
3. Let  $\mathcal{P}(A)$  denote the power set of set  $A$ . Prove that the binary relation  $\subseteq$  on  $\mathcal{P}(A)$  is a partial order.
4. Let  $B = \{0, 1\}$  and define the binary relation  $R$  on  $B \times B$  as follows:  
$$\forall (a, b), (c, d) \in B \times B ((a, b) R (c, d) \Leftrightarrow (a \leq c) \wedge (b \leq d)).$$
  
(a) Prove that  $R$  is a partial order.  
(b) Draw the Hasse diagram for  $R$ .  
(c) Find the maximal, largest, minimal and smallest elements.  
(d) Is  $(B \times B, R)$  well-ordered?

5. Let  $R$  be a binary relation on a non-empty set  $A$ . Let  $x, y \in A$ . Define a relation  $S$  on  $A$  by

$$x S y \Leftrightarrow (x = y) \vee (x R y) \text{ for all } x, y \in A.$$

Show that:

- (a)  $S$  is reflexive;
- (b)  $R \subseteq S$ ; and
- (c) if  $S'$  is another reflexive relation on  $A$  and  $R \subseteq S'$ , then  $S \subseteq S'$ .

What is this relation  $S$  called? (Hint: Refer to Transitive Closure in Lecture 6).

6. Let  $R$  be a binary relation on a set  $A$ .

We have defined antisymmetry in class:  $R$  is **antisymmetric** iff

$$\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y).$$

We define asymmetry here.  $R$  is **asymmetric** iff

$$\forall x, y \in A (x R y \Rightarrow y \not R x).$$

- (a) Find a binary relation on  $A$  that is both asymmetric and antisymmetric.
- (b) Find a binary relation on  $A$  that is not asymmetric but antisymmetric.
- (c) Find a binary relation on  $A$  that is asymmetric but not antisymmetric.
- (d) Find a binary relation on  $A$  that is neither asymmetric nor antisymmetric.

7. **Definitions.** Consider a partial order  $\leq$  on a set  $A$  and let  $a, b \in A$ .

- We say  $a, b$  are **comparable** iff  $a \leq b$  or  $b \leq a$ .
- We say  $a, b$  are **compatible** iff there exists  $c \in A$  such that  $a \leq c$  and  $b \leq c$ .

In question 2, you are given the “divides” relation on  $A = \{1, 2, 4, 5, 10, 15, 20\}$ . List out the pairs of distinct elements in  $A$  that are (a) comparable; (b) compatible. Use the notation  $\{x, y\}$  to represent the pair of elements  $x$  and  $y$ .

8. Let  $A = \{a, b, c, d\}$ . Consider the following partial order on  $A$ :

$$R = \{(a, a), (a, b), (a, c), (a, d), (b, b), (b, c), (c, c), (d, d)\}.$$

- (a) Draw a Hasse diagram of  $R$ .
- (b) Draw Hasse diagrams of all the linearizations of  $R$ .

9. [AY2022/23 Semester 1 Mid-term Test]

Let  $A = \{2, 3, 5, 7, 21, 30, 84, 99\}$  and let  $\leq$  be a partial order on the set  $A$  defined by the “divides” relation, that is,  $x \leq y \Leftrightarrow x|y$ . Which of the following statements are true?

- (i) The partial order has a linearization  $\leq^*$  such that  $21 \leq^* 7$ .
- (ii) The partial order has a linearization  $\leq^*$  such that  $3 \leq^* 2$ .
- (iii) The partial order has a linearization  $\leq^*$  such that  $21 \leq^* 5 \leq^* 84$ .
- (iv) The partial order has a linearization  $\leq^*$  such that  $99 \leq^* 84 \leq^* 30$ .

10. [AY2022/23 Semester 2 Mid-term Test]

Let  $\leq$  be a partial order on a non-empty set  $A$ . A subset  $C$  of  $A$  is called a **chain** if and only if every pair of elements in  $C$  is comparable, that is,  $\forall a, b \in C (a \leq b \vee b \leq a)$ . A **maximal chain** is a chain  $M$  such that  $t \notin M \Rightarrow M \cup \{t\}$  is not a chain. The length of a chain is one less than the number of elements in it.

- (a) Let  $A = \{a, b, c, d\}$  and  $(\mathcal{P}(A), \subseteq)$  be a poset on  $\mathcal{P}(A)$ , where  $\mathcal{P}(A)$  denotes the power set of  $A$ . Write out two maximal chains in  $(\mathcal{P}(A), \subseteq)$ .
- (b) Let  $B = \{2, 3, 5, 6, 7, 11, 12, 35, 385\}$  and  $(B, |)$  be a poset on  $B$ , where  $|$  denotes the divides relation. Draw the Hasse diagram and write out two maximal chains of different lengths in  $(B, |)$ .

11. For each of the following statements, state whether it is true or false and justify your answer.

- (a) In all partially ordered sets, any two comparable elements are compatible.
- (b) In all partially ordered sets, any two compatible elements are comparable.